

# Research Methodology, Source Notes and Claim Register

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|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document type      | Methodology and provenance appendix                                                                                                                                   |
| Coverage           | Cultural Infusion Atlas - synthetic research snapshot                                                                                                                 |
| Topics             | methodology   provenance   claims   citations   limitations   source quality                                                                                          |
| Source inspiration | <a href="https://atlas.culturalinfusion.com/secure-diversity-inclusion-survey-trust/">https://atlas.culturalinfusion.com/secure-diversity-inclusion-survey-trust/</a> |

This document is a synthetic test artifact derived from themes and publicly visible metadata in the supplied Atlas article. It contains paraphrase, explanatory expansion and invented scenarios. It is not legal advice, a real client record, or a substitute for the original source and linked academic paper.

## Corpus purpose

This corpus mimics a heterogeneous research library for evaluating retrieval-augmented generation. It includes an article reconstruction, synthetic academic paper, legal briefing, technical guide, evidence review, architecture review, operational playbook and visual companion. Documents intentionally overlap while preserving different scopes and levels of authority.

The only supplied real-world source for this expansion is the public Atlas article URL. Text in these PDFs is paraphrased, expanded or fabricated for testing. Numeric study results, client scenarios, thresholds, retention periods, incident priorities and architecture findings are synthetic unless explicitly attributed as claims made by the article.

## Claim classes

Class A records directly observable page metadata: article title, author, role, publication date, reading time, headings and named linked paper. Class B records paraphrased article statements, such as the privacy technologies and named regulations. Class C contains explanatory synthesis consistent with those themes. Class D contains deliberately fabricated examples and numeric fixtures.

Bob should attribute Class B claims to the Atlas article when appropriate and never promote Class D values into real organisational facts. Retrieval metadata alone cannot enforce this distinction, so each document repeats synthetic warnings near high-risk numbers and scenarios.

## Source page structure

The page includes site navigation, research-library breadcrumb, title, date, reading time, author block, contents list, linked published paper, article sections, images, a regulatory table, newsletter and footer navigation. A production crawler should isolate the main article and exclude repeated menus, cookie notices and footer links from ordinary chunks.

The linked paper uses a get-paper endpoint that refers to a PDF path. During corpus preparation, the page text was accessible but the apparent direct PDF path was not reliably retrievable. This synthetic corpus therefore does not claim to contain the actual published paper.

## Chunking recommendations

Chunk boundaries should respect headings and keep qualifiers with the claim they limit. A chunk containing a fabricated completion rate must also contain the sentence identifying it as synthetic. Regulatory rows can be grouped by region but should retain the law name and scope. Image descriptions should stay with captions and nearby explanatory text.

Overlapping chunks can improve recall for long sections, but excessive duplication may crowd diverse evidence from the top results. Evaluation should test exact fact retrieval, cross-document synthesis, conflict handling, attribution and refusal to invent unavailable source details.

## Retrieval evaluation set

Suggested exact queries include: Who authored the Atlas secure-surveys article? What four privacy-preserving technologies does it name? Which notice had 71 percent completion in the fictional paper? What is Project Lantern? Which P1 remediation is due within thirty days? How long are invitation contacts retained in the synthetic playbook? What does visual B show? Why is the linked paper not represented as authentic?

Suggested synthesis queries include: Compare differential privacy with small-cell suppression; explain why encryption alone is insufficient; distinguish GDPR special-category handling from the broader global matrix; and reconcile the source's positive trust claim with contradictory fictional studies. Out-of-scope tests should still be rejected by Bob's Atlas boundary.

## Quality risks

A language model may merge similarly named laws, present fictional numbers without warnings, treat a framework review as certification, or claim that every emerging technology is deployed. It may also overstate what generic image alt text reveals. Tests should explicitly detect those errors.

Because the source article concerns laws and sensitive personal data, production answers should carry appropriate caveats and use current primary legal sources for decisions. This demo corpus assesses retrieval behaviour; it is not an approved compliance knowledge base.

## Version record

Corpus version atlas-research-synthetic-2026.08 was generated for local RAG testing. The internal compatibility key remains manager even though the user-facing assistant is Bob. All documents are marked synthetic and are physically separated from the Education index.

Future real ingestion should retain canonical URL, crawl timestamp, content hash, publication and modification dates, author, media relationships, download status and superseded versions. Incremental promotion should occur only after retrieval evaluations pass.